Dr. Jhilam Sinha

Assistant Professor,
Department of Civil Engineering,
Indian Institute of Technology (IIT) Guwahati,
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Assam, India

Personal websites:
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https://www.unsw.edu.au/hdr/jhilam-sinha

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https://www.linkedin.com/in/jhilam-sinha-phd-8a50b2b9/
https://scholar.google.com/citations?user=zF3zl8kAAAAJ&hl=en
https://www.researchgate.net/profile/Jhilam-Sinha/research

Education

Degree	Institute	Area	Year (Month)
Doctor of Philosophy	University of New South Wales, Sydney, Australia	Civil and Environmental Engineering (Water Engineering)	2024 (July)
Master of Technology	Indian Institute of Technology, Guwahati, India	Civil Engineering (Water Resources Engineering and Management)	2018 (June)
Bachelor of Technology	Tezpur University, India	Civil Engineering	2016 (June)

Research

Research Interests

Surface and Vadose Zone hydrology

Hydro-climatology

Reservoir Inflow and Streamflow Forecasting

Hydrological Modelling and Remote Sensing

Water Resources management and Sustainability

Socio-hydrology

Peer Reviewed Publications

1. ***Sinha, J.,*** *Goel, N.K. (2026). A global comparison of skin and near-surface air temperatures reveals contrasting patterns across latitudes.* ***(in communication)***.

2. *Kumar, J., Goel, N.K.,* ***Sinha, J.,*** *Arora, M. (2026) Future-Informed Residual Learning for Himalayan Inflow Forecasting: A Hybrid Physical–Deep Learning Approach* ***(in communication)***.

3. ***Sinha, J.****, Sharma, A., & Marshall, L. (2025). Spatial disaggregation of SMAP soil moisture using higher resolution precipitation data.* ***(under revision)***.

4. **Sinha, J.**, Sharma, A., & Marshall, L. (2026). Reconstruction of soil moisture time series with characterisation of drydown attributes. *Journal of Hydrology*, 135538.

5. Hilly, J. J., **Sinha, J.**, Mani, F. S., Turagabeci, A., Jagals, P., Thomas, D. S. G., ... & Dansie, A. P. (2025). PM2. 5 and PM10 concentrations in urban and peri-urban environments of two Pacific Island Countries. *Atmospheric Pollution Research*, 102454.

6. **Sinha, J.**, Sharma, A., Marshall, L., & Kim, S. (2024). Characterizing satellite soil moisture drydown: A Bivariate Filtering Approach. *Water Resources Research*, *60*(7), e2022WR034019.

7. **Sinha, J.**, Sharma, A., & Marshall, L. (2023). Hydrological sustainability of international virtual water trade. *Environmental Research Letters*, *18*(12), 124037.

8. Kumar, N., **Sinha, J.**, Madramootoo, C. A., & Goyal, M. K. (2020). Quantifying groundwater sensitivity and resilience over peninsular India. *Hydrological Processes*, *34*(26), 5327-5339.

9. **Sinha, J.**, Das, J., Jha, S., & Goyal, M. K. (2020). Analysing model disparity in diagnosing the climatic and human stresses on runoff variability over India. *Journal of Hydrology*, *581*, 124407.

10. **Sinha, J.**, Jha, S., & Goyal, M. K. (2019). Influences of Watershed Characteristics on Long-Term Annual and Intra-Annual Water Balances over India. *Journal of Hydrology, 577*, 123970.

11. **Sinha, J.**, Sharma, A., Khan, M., & Goyal, M. K. (2018). Assessment of the impacts of climatic variability and anthropogenic stress on hydrologic resilience to warming shifts in Peninsular India. *Scientific reports*, *8*(1), 13833.

Conference Papers

1. **Sinha, J.** (2019). Sensitivity of Annual Runoff to Precipitation in Madhya Pradesh, India. 34th *MP Young Scientist Congress (MPYSC), 28 February-1 March 2019, M.P. Council of Science and Technology, Bhopal.*

2. **Sinha, J.**, Hinge, G. (2018). Spatiotemporal trends of PM2.5 concentrations using satellite data in Meghalaya, India. *3rd Indian International conference on air quality monitoring (IICAQM), 6-7 December 2018, IIT madras.*

Conference Presentations

1. Sharma, A., **Sinha, J.**, & Marshall, L. (2025). A Physics Based Satellite Soil Moisture Reconstruction Algorithm. *European Geosciences Union General Assembly (EGU-2025), 27 Apr–2 May 2025, Vienna, Austria.*

2. **Sinha, J.**, Sharma, A. & Marshall, L. (2024). Spatially downscaling SMAP Level 3 soil moisture data incorporating higher resolution precipitation information. *Asia Oceania Geosciences Society (AOGS-2024), 23 June – 28 June 2024, South Korea.*

3. **Sinha, J.**, Sharma, A. & Marshall, L. (2023). Evaluating drydown patterns from SMAP L3 using a bivariate recursive filter. *Asia Oceania Geosciences Society (AOGS-2023), 30 July - 4 August 2023, Singapore.*

4. **Sinha, J.**, Sharma, A., Marshall, L & Kim, S. (2022). Characterising Satellite Soil Moisture Drying Rates Using a Bivariate Recursive Filtering Approach. *American Geophysical Union Fall Meeting (AGU-2022), 12-16 December 2022, Chicago, Illinois.*

5. **Sinha, J.**, Sharma, A., Marshall, L & Kim, S. (2022). Characterisation of satellite soil moisture drydowns using a bivariate filter. *International Conference on Water and Environmental Engineering (ICWEE-2022), 27-30 November 2022, Sydney.*

6. **Sinha, J.**, Sharma, A., Marshall, L & Kim, S. (2021). A bivariate filter for characterization of drying rates of satellite soil moisture drydowns. *Modelling and Simulation (MODSIM-2021), 5-10 December 2021, Sydney.*

Conference Posters

1. **Sinha, J.**, Sharma, A. & Marshall, L. (2025). Reconstruction of Satellite Soil Moisture Series using

Physical Processes. *International Association of Hydrological Sciences Scientific Assembly (IAHS 2025), 05-10 October 2025, Roorkee, India.*

2. **Sinha, J.**, Sharma, A. & Marshall, L. (2022). Sustainability of Virtual Water Trade. *American Geophysical Union Fall Meeting (AGU-2022), 12-16 December 2022, Chicago, Illinois.*

3. **Sinha, J.**, Goyal, M.K. (2019). Influences of local factors on long-term annual and intra-annual water balances across 25 major river basins in India. *World Environmental and Water Resources Congress (EWRI-2019), 19-23 May 2019, Pittsburgh, Pennsylvania.*

Talks

1. **Sinha, J.** (2019). Hydrologic resilience. *Recent advances in Water Resources and Environmental Engineering, 22-27 April 2019, Technical Education Quality Improvement Programme (TEQIP), IIT Indore.*

2. ***Sinha, J.*** *(2026).* Real-time inflow forecast system for Chamera III Dam. *National Hydroelectric Power Corporation (NHPC) Limited office, Faridabad, India.*

Software

1. Precision crop water requirement and dynamic irrigation scheduling tool – The software performs bias correction of soil moisture time series before using the corrected data to estimate crop water requirements and support irrigation scheduling. Users can perform analyses for any location globally. ***(under development)***

Publication Reviewer

Reviewer for Journal of Hydrology

Reviewer for Environmental Research Letters

Reviewer for International Journal of Digital Earth

Reviewer for Theoretical and Applied Climatology

Reviewer for Scientific Reports

Reviewer for Scientific Data

Reviewer for Pure and Applied Geophysics

Reviewer for Regional Environmental Change

Reviewer for Discover Environment

Reviewer for Discover Geoscience

Reviewer for Discover Sustainability

Reviewer for Discover Artificial Intelligence

Services

Session assistant for Asia Oceania Geosciences Society (AOGS-2023) conference in Singapore.

Session assistant for Asia Oceania Geosciences Society (AOGS-2024) conference in South Korea.

Media Highlight

India Water Portal

http://www.indiawaterportal.org/articles/human-activities-reduce-catchments-climate-resilience

Research Matters

https://researchmatters.in/news/india%E2%80%99s-river-catchments-are-drying-too-many-too-soon

Work Experiences

Post-Doctoral Project Fellow in Indian Institute of Technology, Roorkee
Dr. Narendra Kumar Goel- *Emeritus Fellow, IIT Roorkee* *Aug 2025-Oct 2026*

Post-Doctoral Research Associate in Indian Institute of Technology, Roorkee
Dr. Narendra Kumar Goel- *Emeritus Fellow, IIT Roorkee* *May 2025-Aug 2025*

Research Assistant in University of New South Wales, Sydney
Dr. Andrew Dansie – *Associate Professor, UNSW Sydney* *Sept-Dec 2023*

Postdoctoral writing fellow in University of New South Wales, Sydney
Dr. Ashish Sharma - *Professor, UNSW Sydney* *July-Dec 2023*

Course Marker in University of New South Wales, Sydney
CVEN 3501: Water Resources Engineering *Feb-May 2023*

Demonstrator (Teaching) in University of New South Wales, Sydney
CVEN 3501: Water Resources Engineering *Feb 2020-May 2023*

Junior Research Fellow in Indian Institute of Technology, Indore
Dr. Manish Kumar Goyal- *Professor, IIT Indore* *Aug 2018-Aug 2019*

Teaching Assistant in Indian Institute of Technology, Guwahati
Hydraulics and hydraulics structures lab *Aug 2016-June 2018*

Research project as a summer intern in Indian Institute of Technology, Kharagpur
Dr. Paramita Bhattacharya- *Associate Professor, IIT kharagpur* *June 2015*

Summer Intern at Nagaon Paper Mill (NPM), Jagiroad, Assam, India *June 2014*

Achievements and Honors

Postdoctoral Writing Fellow, 2023: For postdoctoral research (post-submission of thesis) at UNSW, Sydney.

Development and Research Training Grant, 2022 (DRTG): For conference during PhD at UNSW, Sydney**.**

University International post graduate award scholarship, 2019 (UIPA): For PhD research at UNSW, Sydney.

B.Tech Gold medalist, 2016: Secured Gold medal (1st Position in graduating batch 2016) in B.tech in Civil Engineering.

M.H.R.D (Ministry of Human Resources Development) GATE Fellowship 2016: For Post-Graduate study at IIT Guwahati

Ishan Vikas summer training, 2015: Selected for Ishan Vikas summer training Program under M.H.R.D (on merit basis) at IIT Kharagpur.

Technical skills

Programming languages: C, C++, MATLAB, and R

Softwares: InVest, ArcMap, ArcSWAT, AutoCAD, and ETABS; **Basic-** HEC-RAS, STAAD Pro V8i, ERDAS, RORB, and ENVI

Membership

American Geophysical Union (AGU)

European Geosciences Union (EGU)

Asia Oceania Geosciences Society (AOGS)

Modeling and Simulation Society of Australia and New Zealand (MSSANZ)